

VAC²: Visual Analysis of Combined Causality in Event Sequences

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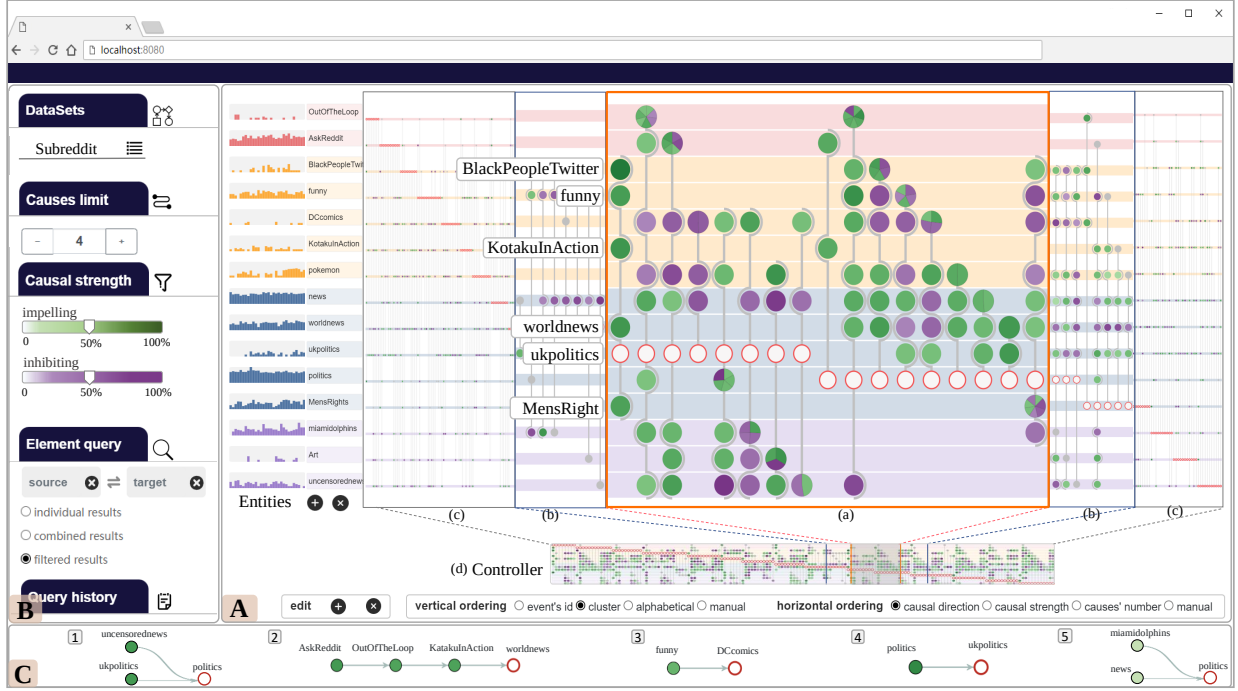


Fig. 1. *User interface*: (A) Combined causality panel for illustrating the individual and combined causes at different level of causality abstraction with controller and diverse ordering. (B) Parameters configuration panel for providing users filtering causality. (C) Propagation panel for finding and tracking causal propagation and storing the history of causal propagation.

Abstract— Identifying causality behind complex systems plays a significant role in different domains, such as decision making, policy implementations, and management recommendations. However, existing causality studies on temporal event sequences data mainly focus on individual causal discovery, which is incapable of exploiting combined causality. To fill the absence of combined causes discovery on temporal event sequence data, eliminating and recruiting principles are defined to balance the effectiveness and controllability on cause combinations. We also leverage the Granger causality algorithm based on the reactive point processes to describe impelling or inhibiting behavior patterns among entities. In addition, we design an informative and aesthetic visual metaphor of “electrocircuit” to encode aggregated causality for ensuring our causality visualization is non-overlapping and non-intersecting. Diverse sorting strategies and aggregation layout are also embedded into our parallel-based, directed and weighted hypergraph for illustrating combined causality. Our developed combined causality visual analysis system can help users effectively explore combined causes as well as an individual cause. This interactive system supports multi-level causality exploration with diverse ordering strategies and a focus and context technique to help users obtain different levels of information abstraction. The usefulness and effectiveness of the system are further evaluated by conducting a pilot user study and two case studies on event sequence data.

Index Terms—Causal discovery, cause combination, directed hypergraph visualization, impelling and inhibiting behaviors, event sequence data

1 INTRODUCTION

Causal discovery mainly focuses on revealing deep and strong correlations among entities. Human activities form a large number of event sequences, where potential causal relations exist among these entity events. Discovering causalities is a ubiquitous and crucial task in many applications such as medical treatment, marketing research, products recommendation, and policy implementation.

In recent years, researchers have been devoting efforts to analyze causal effects [4] and causality problems [18]. Most causality researchers focus on the individual causality discovery and pay little

attention to combined causality discovery. Multifaceted factors are deemed/considered as the a combined cause. An scenario/effect are commonly caused by confounding mutual impacts. For example, High temperature, oxygen, and wood together result in fire, and none of them can cause fire individually. Individual causality discovery obtain result that high temperature, oxygen, and wood be the causes, respectively. Because these researches are conducted under a assumption of neglecting all the other necessary factors. Existing approaches [20] on combined causes discovery can only be applied to multi-dimensional datasets with constraint-based algorithms. No existing algorithms are proposed to discover and analyze combined causal factors for temporal event sequence dataset. In addition, little attention are attracted to detailed description of causality, such as impelling or inhibiting behavior that increase or decrease the occurrence possibility of a certain event.

Meanwhile, many researchers on causality visualization have devoted to simple directed graph layout for efficiently and intuitively

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